Computational Statistics
Unit 5: Integration

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1. Numerical Integration

- In Probability and Statistics integration is very important
- Example: $\mathbb{E}(g(X)) = \int g(x)f(x)dx$
- This is particularly true if we use a bayesian methods, and some optimisation processes such as the EM-algorithm

1. Numerical Integration

Scipy

Introduction to SciPy

SciPy is a Python library for scientific and technical computing, built on top of **NumPy**.

Key Features of SciPy:

- Optimization: `scipy.optimize`
- Integration: `scipy.integrate`
- Interpolation: `scipy.interpolate`
- Linear algebra: `scipy.linalg`
- Signal processing: `scipy.signal`
- Sparse matrices: `scipy.sparse`

SciPy works fine with Numpy. Indeed, Scipy is a sort of specialised version of Numpy. Numpy is good for general manipulation of arrays (like R style arrays), Scipy is more specific.

Scipy vs. Numpy

Numpy is the core package for numerical computing, while **Scipy** builds on top of it to provide advanced scientific algorithms.

Key Differences:

- **Purpose:**
  - `NumPy`: Focused on array manipulation and basic mathematical operations.
  - `SciPy`: Provides advanced mathematical functions and algorithms for optimisation, integration, etc.
- **Functionality:**
  - `NumPy`: Primarily concerned with array operations, linear algebra, and basic statistics.
  - `SciPy`: Offers high-level functionality like `scipy.integrate`, `scipy.optimize`, `scipy.linalg`, among other

In Short: SciPy is built on top of NumPy and offers more specialised algorithms for complex computations.

1. Numerical Integration

Numerical Integration in dimension 1

1 dimensional numerical integration in R

- in Python we can use the modeule scipy.integrate to get different integration methods

```
import numpy as np
import scipy.integrate as si

def f(x):
        return (np.cos(x) + x) * np.exp(-x)

I, error = si.quad(f, 2, np.inf)

print(I)
print(error)
```

- Numerical integration has been implemented in most standard programming languages and scientific libraries (such as scipy).

1. Numerical Integration

Numerical Integration in higher dimension

- Integration in high dimensions is hard.
- Standard numerical methods work well in dimension 1, 2 and maybe 3
- Most methods tend to suffer from the so-called "curse of dimensionality", their complexity depends on the dimension of the problem
- There are a few methods that do not suffer from the curse of dimensionality, among them Monte Carlo Methods

1. Numerical Integration

Numerical Integration in dimension 2 and 3

Dimensions 2 and 3

- The same ideas for 1 dimension integrals can be extended to more dimensions

- In `scipy.integrate` there are function to integrate in higher dimensions such as `dblquad`, `tplquad` for integrating in dimension 2 and 3 respectively, and `nquad` for general integration in dimension n

```
import numpy as np
import scipy.integrate as si

def fun(y, x):
        return np.cos(x) * np.cos(y)

result, error = si.dblquad(fun, 0, 1, lambda x: 0, lambda x: 1)

print(result)
```

In this code, `lambda x` can be a function such as `np.sqrt(x)`.

2. Monte Carlo Method

Introduction

- Monte Carlo methods are a very slow integration method, but it does not get worse when the dimension of the problem grows.
- Monte Carlo method eventually gets better at integrating that standard methods
- High dimension $\approx$ 4 or more
- The idea of Monte Carlo Methods is the Law of the Large Numbers

2. Monte Carlo Method

The Law of the Large Numbers

The Law of the Large Numbers

Theorem (Law of Large Numbers)

Let $(X_i)_{i\geq 1}$ be an i.i.d. sequence of random variables and a density f taking values on $\mathbb{R}^d$. Consider a function $h : \mathbb{R}^d \to \mathbb{R}$ and suppose that $\mathbb{E}(|h(X_1)|) < \infty$, then

$$\frac{1}{n}\sum_{i=1}^{n} h(X_i) \overset{a.s.}{\to} \mathbb{E}(h(X_1)) = \int_{\mathbb{R}^d} h(x)f(x)dx$$

- The actual theorem is way more general
- The theorem essentially states that the empirical average from i.i.d. samples of a population converges to the theoretical mean.
- Since the expectation of a continuous random variable is written as the integral of the density we can guess how to perform a Monte Carlo method.

2. Monte Carlo Method

Monte Carlo Method

Monte Carlo Method

- Suppose we want to compute the integral of a function $g : \mathbb{R}^d \to \mathbb{R}$ in the region $A \subseteq \mathbb{R}^d$

$$\int_A g(x)dx = \int_{\mathbb{R}^d} g(x)\mathbb{1}_A(x)dx$$

- we now choose a density f on $\mathbb{R}^d$ such that $A \subseteq \text{supp}(f)$, and consider

$$\int_{\mathbb{R}^d} \frac{g(x)}{f(x)}\mathbb{1}_A(x)f(x)dx$$

- the integral above can be written as $\mathbb{E}\left(\frac{g(X)}{f(X)}\mathbb{1}_A(X)\right)$ with $X \sim f$.

- Here we can interpretate $0/0 = 0$ as we do not care what happen outside A

Monte Carlo Method

- So far we have
$$\int_A g(x)dx = \mathbb{E}(g(X)/f(X)\mathbb{1}_A(X))$$
- Suppose we know how to sample from f. Let $(X_1, \ldots, X_n)$ a i.i.d. sample from f
- Then by the Law of Large Numbers
$$\mathbb{E}(g(X)/f(X)\mathbb{1}_A(X)) \approx \frac{1}{n}\sum_{i=1}^{n} \frac{g(X_i)}{f(X_i)}\mathbb{1}_A(X_i)$$
- Finding the right density f is a sort of 'art'
- Sometimes the problem guides helps us

Monte Carlo Method: Example 1

Example: Suppose we want to calculate

$$\int_2^\infty (x + \cos(x))e^{-x}dx$$

- By looking at the problem, the density $f(x) = e^{-x}$ defined in $[0, \infty)$ seems to be a good choice.
- This is the density of an $\text{Exp}(1)$ random variable.
- In our case we have

$$\begin{aligned} g(x) &= (x + \cos(x))e^{-x} \\ f(x) &= e^{-x} \\ A &= [2, \infty) \end{aligned}$$

Monte Carlo Method: Example 1

- Let $X \sim \text{Exp}(1)$, then we just need to compute

$$\mathbb{E}\left((X + \cos(X))\mathbb{1}_{\{X \geq 2\}}\right)$$

- Let $X_1, \ldots, X_n$ be an i.i.d. sample of exponential random variables of parameter 1, then

$$\frac{1}{n}\sum_{i=1}^{n}(X_i + \cos(X_i))\mathbb{1}_{\{X_i \geq 2\}} \to \mathbb{E}\left((X + \cos(X))\mathbb{1}_{\{X \geq 2\}}\right)$$

Monte Carlo Method: Example 1

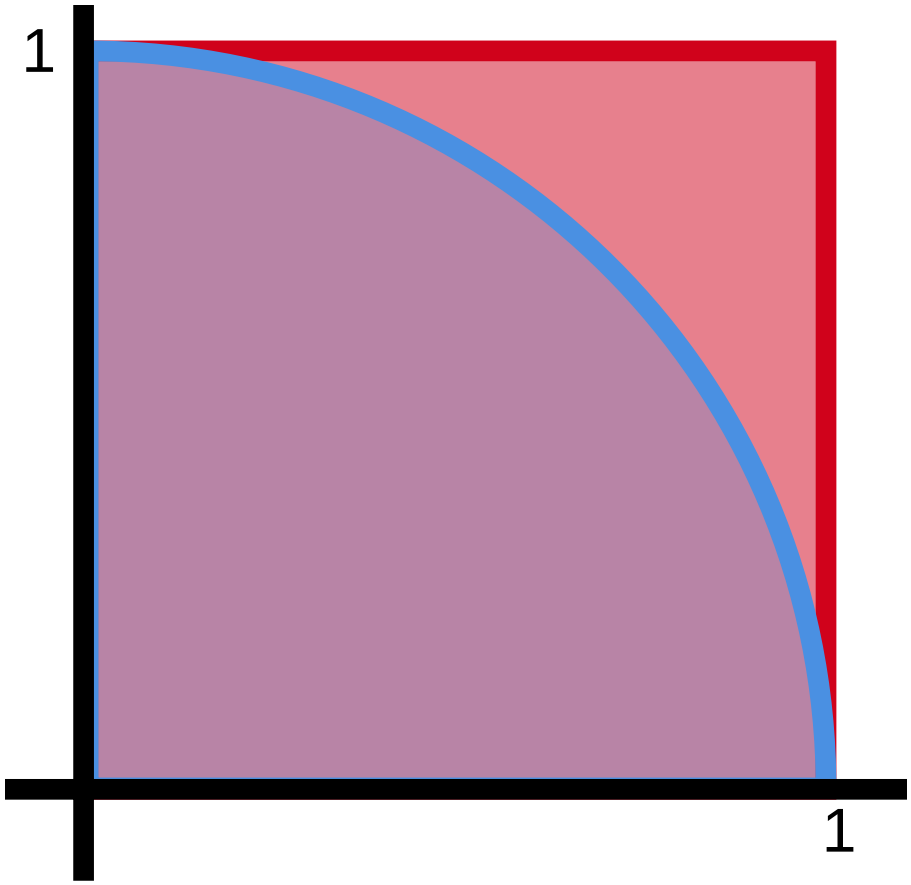
```
import numpy as np
import scipy.integrate as si
import time

def f(x):
        return (np.cos(x) + x) * np.exp(-x)

def h(x):
        return (x + np.cos(x)) * (x >= 2)

def monte_carlo(n):
        sample = np.random.exponential(scale=1.0, size=n)
        return np.sum(h(sample)) / n

start1 = time.time()
I1, error = si.quad(f, 2, np.inf)
end1 = time.time()

start2 = time.time()
I2 = monte_carlo(1000000)
end2 = time.time()

print("Integration result:", I1)
print("Monte Carlo result:", I2)
print("Integration time:", end1 - start1, "seconds")
print("Monte Carlo time:", end2 - start2, "seconds")
```

Monte Carlo Method

- Compared to standard numerical methods, there are no error guarantees for Monte Carlo approximations, since it is random...
- However, we can theoretically compute the variance of the estimator
- Note that Simulation techniques are fundamental to implement montecarlo methods.
- We can also use the Monte Carlo Method for sums (using discrete random variables in this case)

Monte Carlo Method: Example 2 - Approximating π

- This is a classic way to compute π.
- It is a very bad way of approximating π, but pretty cute
- Consider a square with corners $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$. And consider the quarter of circle with radius 1 centred at $(0,0)$.



Monte Carlo Method: Example 2 - Approximating π

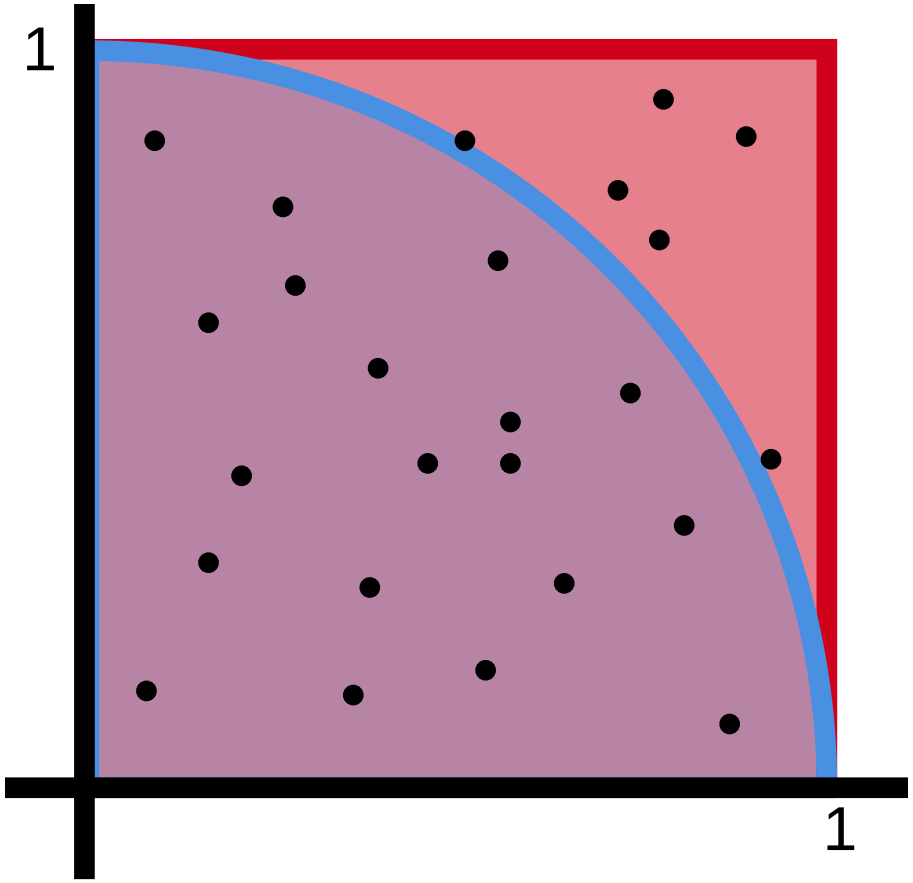
- The area of the square is 1
- The area of the quarter circle is $\pi/4$
- Let B denotes the region enclosed by the quarter circle.
  $B = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x^2 + y^2 \leq 1\}$
- Then

$$\text{Area Circle} = \int_0^1 \int_0^1 \mathbb{1}_B(x, y) dxdy$$

- Consider (U, U') being i.i.d. $\mathcal{U}(0, 1)$. Then

$$\mathbb{E}(\mathbb{1}_B(U, U')) = \int_0^1 \int_0^1 \mathbb{1}_B(x, y) dxdy$$

- In a picture, the idea is



Monte Carlo Method: Example 2 - Approximating π

```python
import numpy as np

n = 1000
points = np.random.uniform(0, 1, size=(n, 2))

# Check which points fall inside the unit circle
inside = np.sum(np.sum(points**2, axis=1) <= 1)

approx_pi = 4 * inside / n

print(approx_pi)
```

Monte Carlo method: Example 3 - An infinitely dimensional problem

- Let $(X_i)_{i\geq 1}$ be an i.i.d. sequence of $\mathrm{Exp}(\lambda)$ random variables.
- Define the random variable N by

$$N = \min\left\{ n \geq 1 : \sum_{i=1}^{n} X_i > 1 \right\}.$$

- **Problem:** Compute $\mathbb{E}(N)$
- $\mathbb{E}(N)$ is a discrete random variable that is a function of continuous-random variables. Then, we should be able to write an appropriate expression using integrals and density functions...
- but we will need to consider integrals in arbitrarily large dimension! (**Why?**)
- For these type of problems either we find a nice theory behind it or we try Monte Carlo approximations
- Exercise: Try to do it by hand

Monte Carlo method: Example 3 - An infinitely dimensional problem

We will need a function to get a sample from N

```
import numpy as np

def sample_N(lambd):
        S = 0.0
        N = 0
        while S < 1:
                S = S+np.random.exponential(scale=1.0 / lambd)
                N = N+1
        return N
```

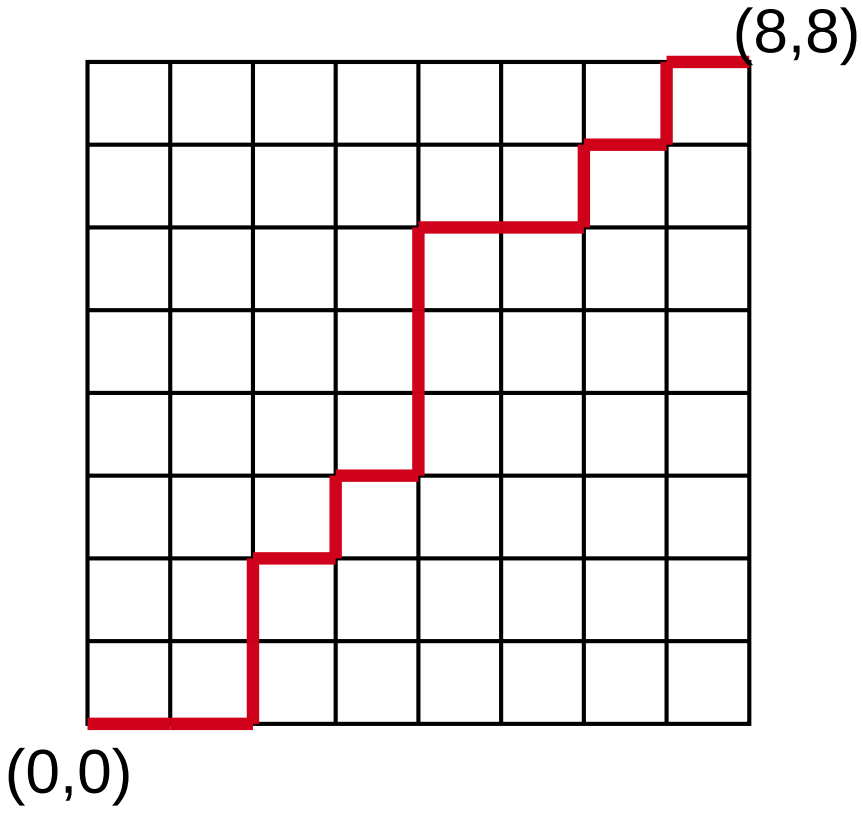
Monte Carlo method: Example 3 - An infinitely dimensional problem

With the function `SampleN` we can write the code for the Monte Carlo method.

```
def monte_carlo_N(n, lambd):
        samples = np.zeros(n)
        for i in range(n):
                samples[i] = sample_N(lambd)
                return np.mean(samples)

n = 1000
lambd = 150
print(monte_carlo_N(n, lambd))
```

Monte Carlo Method: Example 4 - Path problems

- Consider, again, the *NE*-paths on the grid on a $n \times n$ grid



- Let $\mathcal{P}$ the set of *NE* paths from $(0, 0)$ to (n, n).
- **Problem:** Compute the number of paths in $\mathcal{P}$ with ℓ corners or less.

Monte Carlo Method: Example 4 - Path problems

- This counting problem seems to be easy: just enumerate all the paths and counts those with ℓ corners or less.

- Let $C_\ell(p)$ the function that returns 1 if the path p has ℓ corners or less; 0 otherwise

- We want to compute then

$$\sum_{p \in \mathcal{P}} C_\ell(p)$$

- The sum is very hard to compute! $\mathcal{P}$ has $\binom{2n}{n}$ paths! and even enumerating all of them will be painful

Monte Carlo Method: Example 4 - Path problems

- Let's do a Monte Carlo solution
- Note that

$$\sum_{p\in\mathcal{P}} C_\ell(p) = \binom{2n}{n}\left(\sum_{p\in\mathcal{P}} \frac{C_\ell(P)}{\binom{2n}{n}}\right)$$

- $g(p) = 1/\binom{2n}{n}$ is the probability mass function a uniformly chosen path in $\mathcal{P}$
- Therefore

$$\binom{2n}{n}\left(\sum_{p\in\mathcal{P}} \frac{C_\ell(p)}{\binom{2n}{n}}\right) = \binom{2n}{n}\left(\sum_{p\in\mathcal{P}} C_\ell(p)g(p)\right) = \binom{2n}{n}\mathbb{E}(C_\ell(P))$$

  where P is a random path from $\mathcal{P}$
- Of course the Law of Large numbers also apply for discrete distributions, so we can apply the Monte Carlo method here.
- The implementation is sort of long, see the file `MonteCarloPaths.py`

A final and very important remark

- Note that `MonteCarloPaths(20,100000,6)` most of the time returns 0 or 1378465
- This is because `MonteCarloPaths(20,100000,6)` always samples 0 or 1 paths with 6 or less corners
- The problem is that a random paths **concentrate** most of its probability mass on paths with more than 6 corners
- Events of very small probability are called **Rare Events**
- **Rare Events** simulation is an important area of Statistical Computation because it has a lot of important applications: e.g. earthquakes
- **It would be better to use another distribution on paths, that has more mass concentrated around paths with few corners.**
- Choosing the right density/probability mass distribution is key for the Monte Carlo method
- We will study a few options to improve on this issue later

3. Variance Reduction

3. Variance Reduction

Variance of Monte Carlo estimators

Variance of the Monte Carlo estimator

- The main idea of the Monte Carlo method is to estimate $\mathbb{E}(h(X))$ by i.i.d. samples based on the law of large numbers

- Given $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{L}(X)$, then

$$\mathbb{E}(h(X)) \approx \frac{1}{n} \sum_{i=1}^{n} h(X_i)$$

- This is justified by the Law of Large number. Indeed, if $\mathbb{E}(|h(X)|) < \infty$, then

$$\frac{1}{n} \sum_{i=1}^{n} h(X_i) \overset{a.s.}{\to} \mathbb{E}(h(X))$$

- Some authors call $\widehat{I} = \frac{1}{n} \sum_{i=1}^{n} h(X_i)$ a **Monte Carlo Estimator** of $\mathbb{E}(h(X))$

- In general, we cannot quantify the error we are making when using our **Monte Carlo Estimator** because it is random.

- Nevertheless, we can use Probability Theory to say good things about Monte Carlo Estimators (with high probability).

Variance of a Monte Carlo Estimators

Theorem

Consider a distribution F and consider the Monte Carlo estimator

$$\widehat{I} = \frac{1}{n}\sum_{i=1}^{n} h(X_i)$$

with $X_1, \ldots, X_n \overset{iid}{\sim} F$.
Then

1. *$\widehat{I}$ is unbiased, i.e., $\mathbb{E}(\widehat{I}) = \mathbb{E}(h(X_1))$*
2. *if $\mathbb{E}(h(X_1)^2) < \infty$, then $(\widehat{I}) = \frac{1}{n}\mathbb{V}ar(h(X_1))$*

Item 2. is very important as it tells us that the larger is n, the smaller is the variance of the estimator, and hence it stays closer to its mean.

Variance of a Monte Carlo Estimators

Probability theory gives us a few tools to quantify why small variance implies that $\widehat{I}$ is close to its mean

1. Chebyshev's inequality: Let $\varepsilon > 0$ be a fixed number, then

$$\mathbb{P}(|\widehat{I} - \mathbb{E}(\widehat{I})| > \varepsilon) \leq \frac{1}{\varepsilon^2}\mathbb{V}ar(\widehat{I}) = \frac{1}{\varepsilon^2 n}\mathbb{V}ar(h(X_1))$$

   There are stronger inequalities, called concentration inequalities that give much precise bounds

2. Central Limit Theorem

$$\sqrt{n}\left(\widehat{I} - \mathbb{E}(\widehat{I})\right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \mathbb{V}ar(h(X_1)))$$

   - This means that when n is large, and we compare $\widehat{I}$ against is mean, we need to scale by $\sqrt{n}$ to see a meaningful value.
   - In other words, the error term is of order $n^{-1/2}$. This error is independent of the dimension!!

3. Variance Reduction

Importance Sampling

Variance Reduction - Importance Sampling

Importance Sampling is another technique that has two uses:

1. Simplify algorithms when sampling from a distribution is hard
2. Reduce Variance (in some cases)

Variance Reduction - Importance Sampling

The idea of Importance Sampling is quite simple.

1. Let X be a continuous random variable with density f in $\mathbb{R}^d$
2. We are interested in estimating $\mathbb{E}(h(X))$ by using a Monte Carlo estimator
3. Let Y be another continuous random variable with density g
4. The trick is the following.

$$\mathbb{E}(h(X)) = \int_{\mathbb{R}^d} h(x)f(x)dx = \int_{\mathbb{R}^d} \frac{h(x)f(x)}{g(x)} g(x)dx = \mathbb{E}\left(\frac{h(Y)f(Y)}{g(Y)}\right)$$

   We careful and avoid dividing by 0!

5. We can use then the following unbiased estimator of $\mathbb{E}(h(X))$

$$\tilde{I} = \frac{1}{n}\sum_{i=1}^{n} \frac{h(Y_i)f(Y_i)}{g(Y_i)}$$

Variance Reduction - Importance Sampling

Proposition

Let $Y_i \overset{iid}{\sim} g$. *Then*

$$\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{h(Y_i)f(Y_i)}{g(Y_i)}\right) = \mathbb{E}(h(X)).$$

and

$$\mathbb{V}ar\left(\frac{1}{n}\sum_{i=1}^{n}\frac{h(Y_i)f(Y_i)}{g(Y_i)}\right) = \frac{1}{n}\left(\int_{\mathbb{R}^d}\frac{h(x)^2 f(x)^2}{g(x)}dx - \mathbb{E}(h(X))^2\right)$$

If we compare with the usual estimator $\widehat{I} = \frac{1}{n}\sum_{i=1}^{n} h(X_i)$ we notice they have the same expected value, but

$$\mathbb{V}ar\left(\widehat{I}\right) - \mathbb{V}ar\left(\tilde{I}\right) = \int_{\mathbb{R}^d} h(x)^2\left(1 - \frac{f(x)}{g(x)}\right)f(x)dx$$

Variance Reduction - Importance Sampling

As we mentioned there are two uses:

- Sampling from f is hard but sampling from g is easy
- If $\int_{\mathbb{R}^d} h(x)^2 \left(1 - \frac{f(x)}{g(x)}\right) f(x)dx > 0$, the new estimator $\tilde{I}$ has smaller variance.
- Here the '*importance*' part of importance sampling is relevant. If g is a density that focuses on the *important* values, it can estimate $\mathbb{E}(h(X))$ much better.

Variance Reduction - importance Sampling: Toy Example 1

Consider the following simple, but useful example. **Example:** Let $X \sim \text{Exp}(1)$. We want to compute

$$\mathbb{E}\left((X + cos(X))\mathbb{1}_{X \geq 2}\right) = \int_2^{\infty} (x + cos(x))e^{-x}dx$$

- Note that if we perform the usual Monte Carlo estimation then the values $X < 2$ do not add to the sum.
- Maybe we can consider a variable that only takes values in $[2, \infty)$ so we gather only useful information for the computation of the integral)
- What if we consider then the random variables $X^{\star} = 2 + X$?
- We need to find the density function of $2 + X$.

Variance Reduction - Important Sampling: Toy Example 1, cont.

- To compute the density $f^\star$ of $X^\star$ we use

$$\mathbb{P}(X^\star \leq x) = \mathbb{P}(2 + X \leq x) = \mathbb{P}(X \leq x - 2) = (1 - e^{x-2})\mathbb{1}_{x \geq 2}$$

  then, its density $f^*(x) = e^{-(x-2)}\mathbb{1}_{x \geq 2}$.
- let $h(x) = (x + \cos(x))\mathbb{1}_{x \geq 2}$, $\widehat{I}$ the usual Monte Carlo estimator, and $\tilde{I}$ the Monte Carlo estimator given by

$$\tilde{I} = \frac{1}{n}\sum_{i=1}^{n} \frac{h(X_i^\star)f(X_i^\star)}{f^\star(X_i^\star)}$$

- then

$$\begin{aligned}
\mathbb{V}ar\left(\widehat{I}\right) - \mathbb{V}ar\left(\tilde{I}\right) &= \frac{1}{n}\int_0^\infty h(x)^2\left(1 - \frac{f(x)}{f^\star(x)}\right)f(x)dx \\
&= \frac{1 - e^{-2}}{n}\int_2^\infty h(x)^2 f(x)dx \\
&= \frac{1 - e^{-2}}{n}\mathbb{E}(h(X)^2) \geq (1 - e^{-2})\mathbb{V}ar(\hat{I})
\end{aligned}$$

  por lo que concluímos que $\mathbb{V}ar(\tilde{I}) \leq e^{-2}\mathbb{V}ar(\hat{I})$. Algo así como 0.23 la varianza del estimador original

Variance Reduction - Importance Sampling

- In general, finding good densities for importance sampling is hard and a sort of ‘art´.
- General ideas are to consider translation and scaling of random variables $X \to X' = aX + b$
- If we know a bit about the problem, we can create an ad-hoc distribution.
- The idea also works for discrete distributions

Variance Reduction - Importance Sampling - Example 3

The following problem is important in quality control, finance, etc.

- let $(X_i)_{i\geq 1}$ be iid normal random variables with mean $\mu < 0$ and variance 1.
- let $S_m = \sum_{i=1}^{m} X_i$
- let $A, B > 0$ and define $M = \min\{m \geq 1 : \text{either } S_m < -A \text{ or } S_m > B\}$
- **Problem:** Estimate $\mathbb{P}(S_M > B)$.

For a solution, we can do standard Monte Carlo by Sampling S_M a lot of times.
The problem is that since $\mu < 0$, it is more likely that we hit $-A$ before B (probably way more likely) so $\{S_M > B\}$ is a rare event.

An Importance Sampling solution is as follows:

1. Let $(X_i^\star)_{i\geq 1} \overset{iid}{\sim} \mathcal{N}(-\mu, 1)$
2. Define $S_m^\star = \sum_{i=1}^m X_i^\star$
3. Let $M^* = \min\{m \geq 1 : \text{either } S_m^\star < -A \text{ or } S_m^\star > B\}$
4. The idea is that $S_{M^\star}^\star$ has more chances to be greater than B, since $X_i^\star$ has expectation $-\mu > 0$.
5. Finally, denote by f_a the density function of a normal random variable with mean a and variance 1, then
6. Then **(exercise)**:

$$\mathbb{E}\left(\mathbb{1}_{\{S_{M^\star}^\star > B\}} \prod_{i=1}^{M^\star} \frac{f_\mu(X_i^\star)}{f_{-\mu}(X_i^\star)}\right) = \mathbb{E}\left(\mathbb{1}_{\{S_M > B\}}\right) = \mathbb{P}(S_M > B)$$

hint: Here, it is very important to notice that $\mathbb{1}_{S_{M^*}^*}$ is a function of $(X_1^*, X_2^*, \ldots)$
Then, for $m \geq 1$ define R_m by

$$R_m = \{x \in \mathbb{R}^m : -A < \sum_{i=1}^{j} x_i < B \text{ for all } j \leq m-1, \text{ and } \sum_{i=1}^{m} x_i > B\}$$

and note that $S_{M^*}^* > B$ is the union of all the (disjoint) R_m.

7. Now, note that $f_\mu(x)/f_{-\mu}(x) = e^{2\mu x}$
8. then

$$\mathbb{E}\left(\mathbb{1}_{\{S^\star_{M^\star} > B\}} \prod_{i=1}^{M^\star} \frac{f_\mu(X_i^\star)}{f_{-\mu}(X_i^\star)}\right) = \mathbb{E}\left(\mathbb{1}_{\{S^\star_{M^\star} > B\}} e^{2\mu S^\star_{M^\star}}\right)$$

9. As a bonus!

$$\begin{aligned}\mathbb{P}(S_M > B) &= \mathbb{E}\left(\mathbb{1}_{\{S^\star_{M^\star} > B\}} e^{2\mu S^\star_{M^\star}}\right) \\ &= \mathbb{E}\left(\mathbb{1}_{\{S^\star_{M^\star} > B\}} e^{2\mu B}\right)\end{aligned}$$

The last step is justified because the random variable only takes positive values hen $S^*_{M^*} \geq B$, and since $\mu < 0$, we have $\mu S^*_{M^*} \leq \mu B$. Now choose $A \to \infty$ and we get

$$\mathbb{P}(S_m \text{ eventually crosses } B) \leq e^{2\mu B}$$

(recall that $\mu < 0$)

4. Extra Material

4. Extra Material

Variance Reduction - Negative Covariance

Variance Reduction- Negative Covariance

A simple way, yet very limited and occasional, to reduce variance of a Monte Carlo estimator is to consider random variables with negative covariance

Definition: Correlation

Given two random variables X and Y, the covariance between X and Y is given by

$$\mathbb{C}ov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

- When X and Y are independent we have $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ so $\mathbb{C}ov(X, Y) = 0$.
- The covariance is important because

$$\mathbb{V}ar(X + Y) = \mathbb{V}ar(X) + \mathbb{V}ar(Y) + 2\mathbb{C}ov(X, Y)$$

Variance Reduction - Negative Covariance: Example 1

Let's consider the following toy example.

Example 1: estimate $\mathbb{E}(U)$ where $U \sim \mathcal{U}(0,1)$.

- For the sake of the example, let's use $2n$ samples in our estimator
- The usual Monte Carlo estimator is $\widehat{I} = \frac{1}{2n}\sum_{i=1}^{2n} U_i$ with $U_i \overset{iid}{\sim} \mathcal{U}(0,1)$.
- The variance of $\widehat{I}$ is $\frac{1}{2n}\mathbb{V}ar(U_1)$
- Consider now the following estimator

$$\tilde{I} = \frac{1}{2n}\sum_{i=1}^{n}(U_i + U_i')$$

  where $U_i \overset{iid}{\sim} \mathcal{U}(0,1)$ and $U_i' = 1 - U_i$
- Clearly this is an unbiased estimator, and $\mathbb{E}(\tilde{I}) = \mathbb{E}(\widetilde{I}) = \mathbb{E}(U_1)$
- but $\mathbb{V}ar(\tilde{I}) = \frac{\mathbb{V}ar(U_1)+\mathbb{C}ov(U_1,1-U_1)}{2n} = 0$
- $\mathbb{V}ar(\tilde{I}) = 0$ is obvious as $U_i + U_i' = 1$

Variance Reduction - Negative Covariance: Example 2

Let's do something slightly more serious

Example 2: estimate $\mathbb{E}(X)$ where $X \sim \text{Exp}(1)$.

- We consider a similar idea. Let $F(x) = 1 - e^{-x}$, then $F^{-1}(y) = -\log(1-y)$
- Let $U_i \overset{iid}{\sim} \mathcal{U}(0,1)$ and $U_i' = 1 - U_i$
- Define $X_i = -\log(1-U_i)$ and $X_i' = -\log(U_i)$. Note that X_i and X_i' are Exp(() 1).
- then

$$\mathbb{V}ar\left(\frac{1}{2n}\sum_{i=1}^{n}(X_i + X_i')\right) = \frac{\mathbb{V}ar(X_i) + \mathbb{C}ov(X_i, X_i')}{2n}$$

- Now: $\mathbb{C}ov(X_i, X_i') < 0$, why?

$$\mathbb{E}(X_i X_i') = \int_0^1 \log(u)\log(1-u)du = 2 - \frac{\pi^2}{6}$$

and $\mathbb{E}(X_i)^2 = 1$ thus $\mathbb{C}ov(X_i, X_i') = 1 - \pi^2/6 \approx -0.645$

Variance Reduction - Negative Covariance

- Most of the times we won't be able to compute the covariance
- How do we know if our variables have negative covariance?

Proposition

Let $g : \mathbb{R} \to \mathbb{R}$ be an non-decreasing function and let $h : \mathbb{R} \to \mathbb{R}$ be a non-increasing function. Then

$$\mathbb{C}ov(g(X), h(X)) \leq 0$$

For the proof the following characterisation of the covariance is quite useful:

My 2nd favourite trick

Let (X', Y') be an independent copy of the pair (X, Y), then

$$\mathbb{C}ov(X, Y) = \frac{1}{2}\mathbb{E}((X - X')(Y - Y'))$$

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- With the previous proposition we can see why taking U and $1 - U$ in the first example works
- Also shows why taking $-\log(U)$ and $-\log(1 - U)$ works in the second example
- With this result, in the worst case, the variance is the same as with i.i.d. samples

Final Remark: Note that even with this technique we will have that

$$\mathbb{V}ar(I) = c/\sqrt{n}$$

for some constant $c > 0$. So unless, $c = 0$, the variance still will decrease at speed $n^{-1/2}$.